

Cluster Architecture: What's Running and Why

A conceptual map of the [k3s/k8s](#) cluster (*ipc4-9*) — organized by the *purpose* each component serves, not by installation order or namespace. Hardware/OS specifics and a quick-reference software component table are in the appendices, kept separate so this body stays about reasoning rather than inventory.

1. Identity & Workload Attestation

SPIRE provides cryptographic identity for workloads, not just network-perimeter trust. Each node attests itself to the SPIRE server using its TPM (`tpm_dev` node attestor) — proof that a workload is genuinely running on specific, attested hardware, not just “something that can reach the cluster network.” SPIRE then issues short-lived SVIDs (SPIFFE Verifiable Identity Documents) that workloads use to authenticate to each other.

The SPIRE server is pinned to *ipc4* specifically, because the TPM DevID CA tooling only lives there. A `signer-unix` sidecar re-signs a trust bundle every 5 minutes; agent pods fail `verify-bundle` and won't start if that pipeline stalls, which is why `bundle-token-age` is one of the more sensitive alerts on this cluster.

This is the foundation zero-trust rests on: everything downstream (mTLS between services, workload-scoped secret access) assumes SPIRE identity is trustworthy.

2. Secrets & Key Management

OpenBao, a [Vault](#) fork, replaces the cluster's previous 3-node Raft HA Vault deployment (decommissioned 2026-08-05). OpenBao's Transit engine auto-unseals the in-cluster instance, eliminating the manual-unseal step that was a real operational hazard under the old Vault setup. OpenBao is also the backing PKI engine (see §6) and generally the place any workload goes for dynamic credentials rather than a hardcoded Secret.

3. Network Security & Segmentation

Cilium is the CNI (replacing k3s's default flannel — `flannel-backend: none` in the server configs). It runs `kube-proxy-replacement=false` over vxlan and enforces [Kubernetes](#) NetworkPolicy via eBPF. Any namespace with a NetworkPolicy needs an explicit `CiliumNetworkPolicy allow-kubelet-probes` alongside it, or kubelet health checks get blocked by the same enforcement that's supposed to be protecting the workload — a real, recurring gotcha on this cluster.

Cilium is also what makes this a genuine microsegmentation exercise rather than a flat pod network: the `netpol-demo` namespace runs three small pods (`client`, `frontend`, `backend`) behind a default-deny NetworkPolicy with one explicit allow rule (`frontend → backend`). It exists purely as a live enforcement probe — routine health checks confirm `client → backend` is blocked and `frontend → backend` succeeds, so a Cilium regression that silently stopped enforcing NetworkPolicy would be caught immediately rather than discovered the next time it actually mattered.

4. Ingress & Load Balancing

The underlying bare-metal hosts have no load-balancer services, so three components stand in for one:

- **kube-vip** floats a VIP (192.168.88.58) across the three control-plane nodes for HA API server access — `k8s-api.home.skeptomai.com` resolves here.
- **MetalLB** (L2/ARP mode) provides LoadBalancer Services from a pool (.240-.250); k3s's built-in ServiceLB is disabled so MetalLB is the only path.
- **Traefik** sits on the MetalLB-assigned .240 address as the actual ingress controller/reverse proxy in front of HTTP(S) services.

5. GitOps & Delivery

Flux continuously reconciles manifests/ from the GitHub repo into cluster state — kube-vip, MetalLB, Cilium NetworkPolicies, and most platform services come back automatically after a rebuild once Flux itself is bootstrapped. Not everything goes through Flux, though: some experiments are applied manually via `kubectl apply` from *omen* (a laptop used for development, at times remote from the cluster LAN and reached over the [tailnet](#)) against the local repo, a deliberate split between “platform, GitOps-managed” and “experiment, hand-applied.”

6. TLS / PKI

cert-manager automates certificate issuance and renewal. The interesting part is *what* signs those certificates: the `vault-pki-issuer` ClusterIssuer keeps its legacy name (renaming it would break existing `Certificate` references) but actually points at `https://openbao.openbao.svc:8200, path: pki_int/sign/home-lab` — i.e., OpenBao is the real intermediate CA now, reached through cert-manager’s Vault-API-compatible issuer type. There’s also a separate `internal-ca-issuer` (self-signed) for cases that don’t need the full chain. This is one of the places where §2’s secret store and §6’s PKI aren’t really separate systems, just separate concerns against the same backend.

7. Storage

Two tiers, chosen per workload:

- **local-path-provisioner** — fast, node-local disk, the default dynamic PV provisioner. No cross-node availability.
- **nfs-subdir-external-provisioner** — shared network storage backed by an NFS export, for anything that needs state visible from more than one node.

8. Observability

Deliberately **not** fully in-cluster: **Prometheus**, **Grafana**, and **Alertmanager** run standalone via Pelagos on *nazgul* (a NAS server — the single most important home server, used to host centralized services independent of the cluster’s own uptime), outside k3s entirely. This is intentional — the cluster is powered down every night on a schedule, and monitoring that lived *inside* the thing it monitors would go blind exactly when startup/shutdown behavior most needs watching. In-cluster, there’s `kube-state-metrics` and a `node-exporter` DaemonSet feeding that external stack, plus Flux’s own controller metrics exposed via a NodePort patch. Alerts route to Pushover; some (SPIRE exporter down, etc.) are expected to fire during the nightly shutdown window and self-clear on the morning startup, which is a real distinction from an alert that indicates an actual fault.

9. Container Runtime

Pelagos is the CRI implementation on every node (`unix:///run/pelagos/cri.sock`) — a custom Rust runtime, not containerd/CRI-O/Docker. This isn’t just an infrastructure choice: the cluster doubles as the primary integration-test environment for Pelagos itself, coordinated with a sister agent/project that builds and releases it. Cluster bugs get filed upstream against Pelagos with the `cluster-origin` label; Pelagos releases get validated against this cluster before being considered done.

10. Virtualization

KubeVirt runs actual VMs (`virt-handler`, `virt-controller`, `virt-api`, `virt-operator`) as Kubernetes-managed workloads, live-migratable across nodes. This lets anything needing full VM isolation, or a non-Linux/non-container guest, sit under the same control plane as everything else rather than requiring a separate hypervisor layer.

Workloads

The platform services above exist to *serve* these — the actual reason the cluster is running:

- **gruesome** — a hosted deployment of a Rust Z-Machine interpreter (plays classic Infocom text adventure games, e.g. Zork I). Real dogfood: built with Pelagos, pushed to the cluster’s own local registry, and run as an ordinary Deployment behind MetalLB — the simplest possible “does the whole stack work end-to-end” workload.
 - **LLM tooling** — [open-webui](#) (chat UI), a self-hosted **web-search** proxy (wraps [DuckDuckGo](#), used as a tool-call target so LLM agents don’t need direct internet egress), and a [Jupyter](#) notebook environment. The actual LLM inference ([Nemotron-3-Super](#) via [vLLM](#)) runs on a separate DGX Spark node reached over the tailnet — adjacent infrastructure, not itself a cluster workload.
 - **Pelagos build jobs** — ephemeral, one-shot k8s Jobs (pinned to *ipc4* for a consistent hostPath mount of the Pelagos binary/store) that run **pelagos build**
 - **pelagos image push** against a cloned repo, the in-cluster mechanism for building and publishing container images without Docker or Kaniko on *omen*.
 - **Authentik** — self-hosted SSO/identity provider for *humans* logging into applications. Distinct from §1’s SPIRE, which is workload-to-workload identity; Authentik is the user-facing auth layer sitting in front of apps.
 - **Demo/experiment apps** ([https-demo](#), [netpol-demo](#), [ipvs-demo](#)) — living examples from the numbered [experiments/](#) directory, kept running rather than torn down so they double as continuous verification of the concepts they demonstrate (TLS via cert-manager, NetworkPolicy enforcement, IPVS kube-proxy mode).
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Appendix A: Hardware & OS Reference

Nodes	<i>ipc4-9</i> , six identical-generation Intel machines
<i>ipc4-6</i>	HP Elite Mini 800 G9, Intel Core i5-12500T (12th Gen), 6c/12t, 32GB RAM, NVMe — role: control-plane + etcd, <code>node-class=performance</code>
<i>ipc7-9</i>	Intel Core i5-12500 (12th Gen, non-T), 6c/12t, 32GB RAM, ~256GB NVMe — role: worker, <code>node-class=fastest</code>
OS	Ubuntu 26.04 LTS, kernel 7.0.x
k3s	v1.35.5
TPM (attestation hardware)	<i>ipc4-7</i> : Nuvoton NPCT75x · <i>ipc8-9</i> : Infineon SLB9672
Container runtime	Pelagos CRI, all nodes
History	<i>ipc1-3</i> (Pentium Gold G5400T) were the original control plane until 2026-07-05, retired and replaced by <i>ipc4-6</i>

Adjacent, not cluster members:

<i>spark-0d93</i>	NVIDIA DGX Spark (GB10 GPU) — standalone, runs vLLM/Nemotron-3-Super via Pelagos (non-CRI mode); reached over Tailscale; wired Ethernet as of 2026-08-19 (was WiFi)
<i>nazgul</i>	NAS server — the single most important home server, host for centralized services; runs Prometheus/Grafana/Alertmanager, the Zot registry mirror/local registry, and other services deliberately kept outside the k3s cluster
<i>omen</i>	Development laptop — at times remote from the cluster LAN, reached via the tailnet; source of <code>kubect1 apply</code> for hand-applied manifests and the origin of the local repo clone

Network:

IP(s)	Purpose
.55-.57	<i>ipc4-6</i> (control-plane), static MikroTik leases by MAC
.58	kube-vip control-plane VIP (<code>k8s-api.home.skeptomai.com</code>)
.63-.65	<i>ipc7-9</i> (workers), static MikroTik leases by MAC
.240-.250	MetalLB LoadBalancer pool

Appendix B: Software Component Reference

Component	Category	Reason for being	Depends on / notes
SPIRE	Identity	TPM-attested workload identity, SVID issuance	TPM hardware (server pinned to <i>ipc4</i>), k8s API
OpenBao	Secrets	Secret storage, Transit auto-unseal, PKI backend	Replaces old 3-node Raft HA Vault
Cilium	Networking	CNI, NetworkPolicy enforcement via eBPF	Replaces flannel; needs <code>allow-kubelet-probes</code> netpol per namespace
kube-vip	Networking	HA control-plane VIP	L2/ARP, <i>ipc4-6</i>
MetalLB	Networking	LoadBalancer IP pool	L2/ARP mode; replaces k3s ServiceLB
Traefik	Networking	Ingress controller / reverse proxy	Sits behind MetalLB .240
Flux	GitOps	Reconciles manifests/ from GitHub	Not all workloads are Flux-managed
cert-manager	PKI	Automated cert issuance/renewal	Issuers backed by OpenBao + self-signed
local-path-provisioner	Storage	Fast node-local dynamic PVs	No cross-node availability
nfs-subdir-external-provisioner	Storage	Shared network storage	Backed by an NFS export
kube-state-metrics / node-exporter	Observability	Cluster/node metrics feed	Scraped by external Prometheus (<i>nazgul</i>)
Prometheus / Grafana / Alertmanager	Observability	Metrics, dashboards, alerting	Runs outside k3s, on <i>nazgul</i> , by design
Pelagos	Container Runtime	CRI implementation (custom Rust, not Docker)	Every node; also the project this cluster validates
KubeVirt	Virtualization	Runs VMs as k8s-managed workloads	virt-handler/controller/api/operator
gruesome	Workload	Hosted Z-Machine interpreter (text adventures)	Built via Pelagos, pushed to local registry
open-webui	Workload	Chat UI for LLM interaction	Talks to spark's vLLM endpoint
web-search	Workload	Self-hosted DuckDuckGo proxy for tool-calling	Used by gptel's <code>web_search</code> tool
jupyter	Workload	Interactive notebook environment	
Authentik	Workload	Human-facing SSO/identity provider	Distinct from SPIRE (workload identity)
https-demo / netpol-demo / ipvs-demo	Workload	Living examples from <code>experiments/</code>	Kept running as continuous verification

Appendix C: Online References

Canonical project source for every named technology in this document, alphabetical.

- **Alertmanager** — <https://github.com/prometheus/alertmanager>
- **Authentik** — <https://github.com/goauthentik/authentik>
- **cert-manager** — <https://github.com/cert-manager/cert-manager>
- **Cilium** — <https://github.com/cilium/cilium>
- **DuckDuckGo** — <https://duckduckgo.com/>
- **Flux** — <https://github.com/fluxcd/flux2>
- **Grafana** — <https://github.com/grafana/grafana>
- **gruesome** — <https://github.com/skeptomai/gruesome>
- **Jupyter** — <https://jupyter.org/>
- **k3s** — <https://github.com/k3s-io/k3s>
- **kube-vip** — <https://github.com/kube-vip/kube-vip>
- **Kubernetes** — <https://github.com/kubernetes/kubernetes>
- **KubeVirt** — <https://github.com/kubevirt/kubevirt>
- **local-path-provisioner** — <https://github.com/rancher/local-path-provisioner>
- **MetalLB** — <https://github.com/metallb/metallb>
- **Nemotron-3-Super** — <https://huggingface.co/nvidia/NVIDIA-Nemotron-3-Super-120B-A12B-NVFP4>
- **nfs-subdir-external-provisioner** — <https://github.com/kubernetes-sigs/nfs-subdir-external-provisioner>
- **open-webui** — <https://github.com/open-webui/open-webui>
- **OpenBao** — <https://github.com/openbao/openbao>
- **Pelagos** — <https://github.com/pelagos-containers/pelagos>
- **Prometheus** — <https://github.com/prometheus/prometheus>
- **SPIRE** — <https://github.com/spiffe/spire>
- **Tailscale** — <https://tailscale.com/>
- **Traefik** — <https://github.com/traefik/traefik>
- **Vault** — <https://github.com/hashicorp/vault>
- **vLLM** — <https://github.com/vllm-project/vllm>
- **Zot** — <https://github.com/project-zot/zot>